

E-Commerce Microservices — Entities & Relationships

1. Auth Service

Entities

User

Fields:

- id
- username
- email
- password
- role
- createdAt

Role

Fields:

- id
- roleName

Relationship

Role (1) -----> (Many) Users

One role can belong to many users.

Example:

- ROLE_USER
 - ROLE_ADMIN
-

2. Product Service

Entities

Product

Fields:

- id
- name
- description
- price
- stockQuantity
- imageUrl
- createdAt
- updatedAt

Category

Fields:

- id
- name
- description

Relationship

Category (1) -----> (Many) Products

One category contains many products.

Example:

- Electronics
- Clothing
- Mobiles

3. Order Service

This is the most important service.

Entities

Cart

Fields:

- id
- userId
- createdAt

CartItem

Fields:

- id
- cartId
- productId
- quantity
- price

Order

Fields:

- id
- userId
- totalAmount
- orderStatus
- paymentStatus
- createdAt

OrderItem

Fields:

- id
- orderId
- productId
- quantity
- price

Relationships

Cart (1) -----> (Many) CartItems

Order (1) -----> (Many) OrderItems

External References

Order Service references:

- userId from Auth Service
- productId from Product Service

DO NOT use JPA entity relationships across microservices.

Use only IDs.

Example:

```
private Long userId;  
private Long productId;
```

4. Payment Service

Entities

Payment

Fields:

- id
- orderId
- amount
- paymentMethod
- paymentStatus
- transactionId
- createdAt

Relationship

Order (1) -----> (1) Payment

Payment belongs to one order.

5. Notification Service

Entities

Initially you can keep this simple.

NotificationLog

Fields:

- id
 - userId
 - message
 - type
 - status
 - createdAt
-

Relationship

References:

- userId from Auth Service

No direct entity relationship needed.

6. Review Service (Optional Strong Add-On)

Entities

Review

Fields:

- id
- userId
- productId
- rating
- comment
- createdAt

Relationship

Product (1) -----> (Many) Reviews

User (1) -----> (Many) Reviews

Again:

- use productId
- use userId

not JPA mapping across services.

7. Analytics Service (Optional)

Entities

ProductAnalytics

Fields:

- id
- productId
- totalOrders
- totalRevenue
- totalViews

SalesAnalytics

Fields:

- id
 - totalRevenue
 - totalOrders
 - analyticsDate
-

Important Microservice Rule

Inside SAME service:

- use JPA relationships

Across DIFFERENT services:

- NEVER use entity mapping
 - communicate via REST/WebClient
 - store only IDs
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Important Internal Relationships

Product Service

Category (1) -----> (Many) Products

Order Service

Cart (1) -----> (Many) CartItems

Order (1) -----> (Many) OrderItems

Recommended Entity Count

Auth Service

- User
- Role

Product Service

- Product
- Category

Order Service

- Cart
- CartItem
- Order
- OrderItem

Payment Service

- Payment

Notification Service

- NotificationLog

Review Service

- Review

Analytics Service

- ProductAnalytics
- SalesAnalytics

Suggested Order Status Enum

PLACED,
PAID,
PROCESSING,
SHIPPED,
DELIVERED,
CANCELLED

Suggested Payment Status Enum

PENDING,
SUCCESS,
FAILED

Suggested Relationships Summary

```
Role → Users  
Category → Products  
Cart → CartItems  
Order → OrderItems  
Order → Payment  
Product → Reviews  
User → Reviews
```

Important Advice

Do NOT overcomplicate entity design initially.

Your goal should be:

- clean architecture
- proper relationships
- good workflows
- strong service communication
- transaction handling

That already creates a strong backend project.